

Last Time

Review: ANOVA Model

Effect Size and Power

Post-hoc Tests

- Familywise vs. Per Comparison

- Bonferroni Inequality

- Range Tests

GSS 1980

In Class Exercise

Today

Tukey's HSD Test

GSS 1980

In Class Exercise

Contrasts

Factorial ANOVA

- Partitioning the Variance

- Simple Effects

- Review of Effect Sizes

Studentized Range Statistic

Let $\{X_1, X_2, \dots, X_N\}$ be a sample from a $\mathcal{N}(\mu, \sigma)$ distribution. Then

$$q = \frac{\max X - \min X}{s}$$

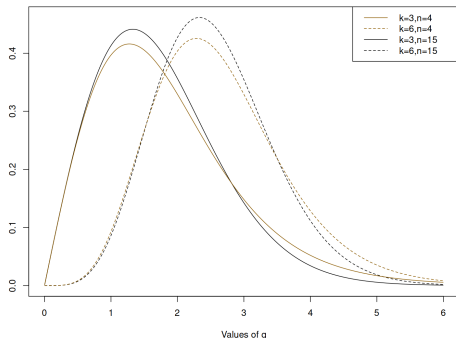
follows the *Studentized range distribution*.

Studentized Range Statistic

Let $\{\bar{X}_1, \bar{X}_2, \dots, \bar{X}_k\}$ be a sample from a $\mathcal{N}(\mu, \sigma/\sqrt{n})$ distribution. Then

$$q = \frac{\bar{X}_{\max} - \bar{X}_{\min}}{s_P/\sqrt{n}}$$

follows the *Studentized range distribution*, where n is the group size.



Tukey's HSD Test

- ▶ Make pairwise comparisons between all means while holding the overall probability of making a Type I error constant.
- ▶ Must determine a critical “honestly significant difference.”
- ▶ Compute the differences between all pairs of means.
- ▶ Any difference that exceeds the critical value is “honestly significant.”

The Honestly Significant Difference

There must be a value of $\bar{X}_{\max} - \bar{X}_{\min}$ that is big enough to reject $H_0 : \mu_1 = \cdots = \mu_k$. This is the Honestly Significant Difference (HSD). Rearrange the expression for q :

$$q_{(\alpha; df, k)} = \frac{HSD}{\sqrt{\frac{MS_W}{n}}}$$

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$$q_{(\alpha;df,k)} = \frac{HSD}{\sqrt{\frac{MS_W}{n}}}$$

$$HSD = q_{(\alpha;df,k)} \sqrt{\frac{MS_W}{n}}$$

MS_W = mean square within from ANOVA

n = number of people in each group

Fear of Heights Review

	1	2	3	4	5
$\bar{X}_j$	40.16	42.45	42.55	39.89	53.50
N_j	20	20	20	20	20

Source	SS	df	MS	F
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

Differences of Means

	40.16	42.45	42.55	39.89	53.5
40.16		-2.29	-2.39	0.27	-13.34
42.45			-0.1	2.56	-11.05
42.55				2.66	-10.95
39.89					-13.61
53.5					

```
> qtuke(.95,5,95) * sqrt(105.3/20)  
[1] 9.023895
```

Tukey's HSD Test

```
> TukeyHSD(results,ordered=TRUE)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = X ~ group)

$group
      diff      lwr      upr    p adj
1-4  0.26870729 -8.756314  9.293729 0.9999892
2-4  2.56023271 -6.464789 11.585254 0.9332594
3-4  2.65714727 -6.367874 11.682169 0.9242538
5-4 13.60815945  4.583138 22.633181 0.0005807
2-1  2.29152542 -6.733496 11.316547 0.9546428
3-1  2.38843998 -6.636582 11.413462 0.9475290
5-1 13.33945216  4.314430 22.364474 0.0007841
3-2  0.09691456 -8.928107  9.121936 0.9999998
5-2 11.04792674  2.022905 20.072948 0.0084451
5-3 10.95101218  1.925990 19.976034 0.0092643

> 1 - ptukey(2.65714727 / sqrt(105.3/20),5,95)
[1] 0.9242218
```

What if the groups are of different sizes?

How do we compute n for the HSD?

What if the groups are of different sizes?

How do we compute n for the HSD?

If there are unequal group n_j s, then use

$$n = \frac{k}{1/n_1 + 1/n_2 + \cdots 1/n_k},$$

the *harmonic mean* of the n_j s.

What if the groups are of different sizes?

How do we compute n for the HSD?

Tukey-Kramer approach: Remember the Welch's t -test.

$$s_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = \sqrt{\frac{s_P^2}{n_1} + \frac{s_P^2}{n_2}}$$
$$\rightarrow \sqrt{\frac{\frac{MS_{\text{Within}}}{n_i} - \frac{MS_{\text{Within}}}{n_j}}{2}}$$

What if the groups are of different sizes?

How do we compute n for the HSD?

Tukey-Kramer approach: Remember the Welch's t -test.

$$\begin{aligned} s_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} &= \sqrt{\frac{s_P^2}{n_1} + \frac{s_P^2}{n_2}} \\ &\rightarrow \sqrt{\frac{\frac{MS_{\text{Within}}}{n_i} - \frac{MS_{\text{Within}}}{n_j}}{2}} \\ &= \sqrt{\frac{MS_{\text{Within}}}{2}} \sqrt{\frac{1}{n_i} + \frac{1}{n_j}} \\ &= \sqrt{MS_{\text{Within}}/n} \quad \text{if } n_i = n_j. \end{aligned}$$

GSS 1980 ANOVA

```
> data <- read.csv("GSS_1980.csv")
> money <- data$REALRINC
> happy <- as.factor(data$HAPPY < 3)
```

```
> gss <- aov(money~happy)
> summary(gss)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
happy	1	3.189e+09	3.189e+09	5.51	0.0191 *
Residuals	871	5.040e+11	5.787e+08		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

GSS 1980 t-test

```
> t.test(money~happy,var.equal=TRUE)
```

Two Sample t-test

```
data: money by happy
t = -2.3473, df = 871, p-value = 0.01913
alternative hypothesis: true difference in means between group FALSE
and group TRUE is not equal to 0
95 percent confidence interval:
 -10877.8729  -970.7736
sample estimates:
mean in group FALSE  mean in group TRUE
      17924.87      23849.19
```

ANOVA vs. t -test

ANOVA ($\alpha = 0.05$)

$$F(1, 871) = 5.51$$

$$F_C(1, 871) = 3.85$$

$$p = 0.019$$

t -test ($\alpha = 0.05$)

$$t(871) = -2.3473$$

$$t_C(871) = \pm 1.9627$$

$$p = 0.0191$$

Therefore,

$$t_{\alpha/2}(df)^2 = F_{\alpha}(1, df) :$$

performing an ANOVA on two groups is mathematically equivalent to performing a *two-tailed* independent samples t test.

Data

Participants rate guilt/innocence on a 10-point scale (1 = certain she is innocent, 10 = certain she is guilty)

Juror	Record	Group		
		Clean	Control	
1	10	5	4	
2	7	1	6	
3	5	3	9	
4	10	7	3	
5	8	4	3	
$\bar{X}_j$	8	4	5	$\bar{\bar{X}} = 5.67$

- ▶ What is the power of this test to detect a medium effect size (η^2 or $\omega^2 = 0.06$)?
- ▶ Compute η^2 and ω^2 .
- ▶ Use the Bonferroni procedure to determine which means differed from the others.
- ▶ Perform Tukey's HSD test to determine which means differed from the others.

A few words about contrasts

A *contrast* is a difference between means, usually expressed as

$$\Psi = \beta_1\mu_1 + \beta_2\mu_2 + \cdots + \beta_k\mu_k,$$

where $\sum_{i=1}^k \beta_i = 0$.

Example: Criminal Record and Jury Perception

	Group		
	Record	Clean	Control
Mean	8	4	5

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	Group		
	Record	Clean	Control
Mean	8	4	5

Hypothesis: Knowledge of previous guilt taints jury perception toward guilty verdicts.

$$H_0 : \mu_1 = \frac{1}{2}(\mu_2 + \mu_3)$$

Example: Criminal Record and Jury Perception

	Group	
	Record	Clean
Mean	8	4
	Control	
	5	

Hypothesis: Knowledge of previous guilt taints jury perception toward guilty verdicts.

$$H_0 : \mu_1 = \frac{1}{2}(\mu_2 + \mu_3)$$

Define

$$\Psi = \mu_1 - \frac{1}{2}(\mu_2 + \mu_3) = \mu_1 - \frac{1}{2}\mu_2 - \frac{1}{2}\mu_3,$$

so $(\beta_1, \beta_2, \beta_3) = (1, -1/2, -1/2)$.

Test of Contrasts

Rewrite

$$H_0 : \mu_1 = \frac{1}{2}(\mu_2 + \mu_3)$$

as

$$H_0 : \Psi = 0$$

Estimate Ψ with

$$\hat{\Psi} = \bar{X}_1 - \frac{1}{2}\bar{X}_2 - \frac{1}{2}\bar{X}_3 = 8 - \frac{1}{2}(4 + 5) = 3.5$$

What to do with a contrast?

- ▶ Can incorporate $\hat{\Psi}$ into the ANOVA table and use F -tests
- ▶ Can perform t -tests
- ▶ Planned vs. post-hoc contrasts

In R

```
> # Note that factors are ordered alphabetically for character data
> levels(data$Group)

[1] "Clean"    "Control"  "Record"

> # Planned contrasts (orthogonal)
> c1 <- c(-.5,-.5,1) # Clean+Control = Record
> c2 <- c(-1,1,0) # Clean = Control
> mat <- cbind(c1,c2)

> # Tell R what the contrasts of interest are
> contrasts(data$Group) <- mat

> # Compute the values of Psi
> means <- aggregate(data$Rating,by=list(data$Group),mean)$x
> Psi <- means%*%mat # Matrix multiplication
> Psi

      c1 c2
[1,] 3.5  1
```

In R

```
> jury.model <- aov(Rating~Group,data=data)
> summary.aov(jury.model,
>             split=list(Group=list("Record=Clean+Control"=1,
>                                   "Clean=Control"=2)))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Group	2	43.33	21.67	4.062	0.0449	*
Group: Record=Clean+Control	1	40.83	40.83	7.656	0.0171	*
Group: Clean=Control	1	2.50	2.50	0.469	0.5066	
Residuals	12	64.00	5.33			

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Definitions

Definition (Factor)

A *factor* is an independent variable in an ANOVA.

Definition (Factorial Design)

A *factorial design* is a design in which there are two or more independent variables and each level of each variable is paired with each level of all the other variables.

Definition (K -way Factorial Design)

An experimental design with K independent variables, and each level of each variable is paired with each level of all the other variables.

Example: Eysenck, 1974

Why do older people seem not to remember things as well as younger people?

	Memory Task				
Age	Counting	Rhyming	Adjective	Imagery	Intentional
Younger					
Older		(Cell)			

5×2 factorial design

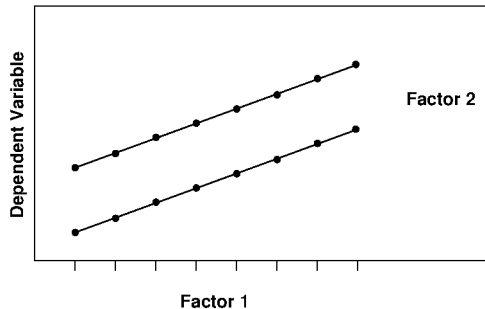
Interactions

Definition (Interaction)

An *interaction* is what happens when the effect of one factor depends on the level of the other factor(s).

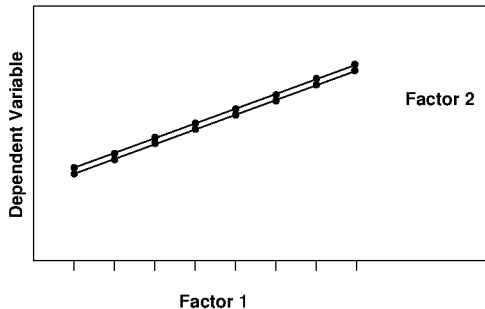
Data Patterns

Plot the means of each cell:



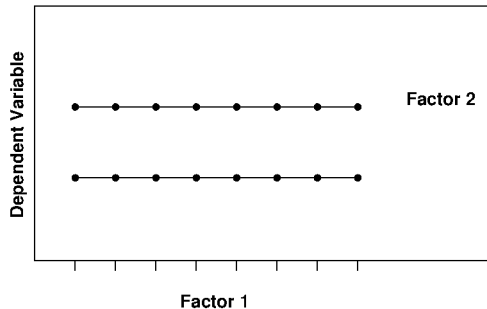
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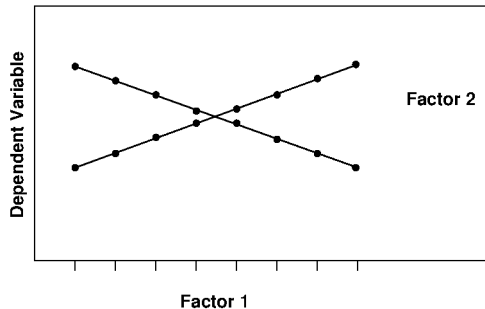
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Data Patterns

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Example: Eysenck, 1974

Why do older people seem not to remember things as well as younger people?

Age	Memory Task					Means
	Counting	Rhyming	Adjective	Imagery	Intentional	
Younger	$\bar{X}_{11}$	$\bar{X}_{12}$	$\bar{X}_{13}$	$\bar{X}_{14}$	$\bar{X}_{15}$	$\bar{X}_{1.}$
Older	$\bar{X}_{21}$	$\bar{X}_{22}$	$\bar{X}_{23}$	$\bar{X}_{24}$	$\bar{X}_{25}$	$\bar{X}_{2.}$
Means	$\bar{X}_{.1}$	$\bar{X}_{.2}$	$\bar{X}_{.3}$	$\bar{X}_{.4}$	$\bar{X}_{.5}$	$\bar{\bar{X}}$

Example: Eysenck, 1974

Why do older people seem not to remember things as well as younger people?

Age	Memory Task					Ns
	Counting	Rhyming	Adjective	Imagery	Intentional	
Younger	N_{11}	N_{12}	N_{13}	N_{14}	N_{15}	$N_{1.}$
Older	N_{21}	N_{22}	N_{23}	N_{24}	N_{25}	$N_{2.}$
Ns	$N_{.1}$	$N_{.2}$	$N_{.3}$	$N_{.4}$	$N_{.5}$	N_{Total}

Example: Eysenck, 1974

Why do older people seem not to remember things as well as younger people?

$$N_{\text{Total}} = 100$$

$$N_{1.} = N_{2.} = 50$$

$$N_{.1} = N_{.2} = N_{.3} = N_{.4} = N_{.5} = 20$$

$$N_{jk} = 10$$

Memory Task						
Age	Counting	Rhyming	Adjective	Imagery	Intentional	Means
Younger	6.5	7.6	14.8	17.6	19.3	13.16
Older	7.0	6.9	11.0	13.4	12.0	10.06
Means	6.75	7.25	12.9	15.5	15.65	11.61

Review: One-way ANOVA

$$SS_{\text{Total}} = \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{\bar{X}})^2$$

$$SS_{\text{Within}} = \sum_{j=1}^k (N_j - 1) s_j^2$$

$$SS_{\text{Between}} = \sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2$$

Two-way ANOVA

$$SS_{\text{Total}} = \sum_{k=1}^K \sum_{j=1}^J \sum_{i=1}^{N_{jk}} (X_{ijk} - \bar{\bar{X}})^2$$

$$SS_{\text{Cells}} = \sum_{k=1}^K \sum_{j=1}^J N_{jk} (\bar{X}_{jk} - \bar{\bar{X}})^2$$

Two-way ANOVA

$$SS_{\text{Total}} = \sum_{k=1}^K \sum_{j=1}^J \sum_{i=1}^{N_{jk}} (X_{ijk} - \bar{\bar{X}})^2$$

$$SS_{\text{Cells}} = \sum_{k=1}^K \sum_{j=1}^J N_{jk} (\bar{X}_{jk} - \bar{\bar{X}})^2$$

$$SS_J = \sum_{j=1}^J N_{j\cdot} (\bar{X}_{j\cdot} - \bar{\bar{X}})^2$$

$$SS_K = \sum_{k=1}^K N_{\cdot k} (\bar{X}_{\cdot k} - \bar{\bar{X}})^2$$

Two-way ANOVA

$$SS_{\text{Total}} = \sum_{k=1}^K \sum_{j=1}^J \sum_{i=1}^{N_{jk}} (X_{ijk} - \bar{\bar{X}})^2$$

$$SS_{\text{Cells}} = \sum_{k=1}^K \sum_{j=1}^J N_{jk} (\bar{X}_{jk} - \bar{\bar{X}})^2$$

$$SS_J = \sum_{j=1}^J N_{j\cdot} (\bar{X}_{j\cdot} - \bar{\bar{X}})^2$$

$$SS_K = \sum_{k=1}^K N_{\cdot k} (\bar{X}_{\cdot k} - \bar{\bar{X}})^2$$

$$SS_{JK} = SS_{\text{Cells}} - SS_J - SS_K$$

$$SS_{\text{Within}} = SS_{\text{Total}} - SS_{\text{Cells}}$$

Example: Eysenck, 1974

Why do older people seem not to remember things as well as younger people?

Memory Task						
Age	Counting	Rhyming	Adjective	Imagery	Intentional	Means
Younger	6.5	7.6	14.8	17.6	19.3	13.16
Older	7.0	6.9	11.0	13.4	12.0	10.06
Means	6.75	7.25	12.9	15.5	15.65	11.61

$$SS_{\text{Total}} = 2667.79$$

$$SS_{\text{Cells}} = 1945.49$$

$$SS_{\text{Age}} = 240.25$$

$$SS_{\text{Task}} = 1514.94$$

$$SS_{\text{Age} \times \text{Task}} =$$

$$SS_{\text{Within}} =$$

ANOVA Summary Table

Source	df	SS	MS	F
Age				
Task				
Age $\times$ Task				
Within				
Total				

In R

```
> eysenck <- aov(recall~task+age+task:age,data=data)
> summary(eysenck)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
task	4	1514.9	378.7	47.191	< 2e-16 ***
age	1	240.3	240.3	29.936	3.98e-07 ***
task:age	4	190.3	47.6	5.928	0.000279 ***
Residuals	90	722.3	8.0		

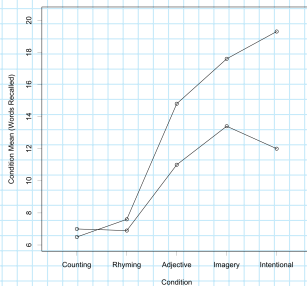
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Simple Effects

Definition (Simple Effect)

A *simple effect* is the effect of a factor measured at a fixed level of another factor.

Eysenck Data



Memory Task

Age	Counting	Rhyming	Adjective	Imagery	Intentional	Means
Younger	6.5	7.6	14.8	17.6	19.3	13.16
Older	7.0	6.9	11.0	13.4	12.0	10.06
Means	6.75	7.25	12.9	15.5	15.65	11.61

Simple Sum of Squares

Consider two factors, J and K . What is the sum of squares for factor J when $K = k$?

$$SS_J \text{ at } K=k = \sum_{j=1}^J N_{jk} (\bar{X}_{jk} - \bar{X}_{\cdot k})^2$$

Simple Sum of Squares

Consider two factors, Age and Task. What is the sum of squares for Age when Task = Intentional?

$$SS_{\text{Age at Intentional}} = \sum_{j=1}^2 N_{j,\text{Intentional}} (\bar{X}_{j,\text{Intentional}} - \bar{X}_{\cdot,\text{Intentional}})^2$$

Simple Sum of Squares

Consider two factors, Age and Task. What is the sum of squares for Age when Task = Intentional?

$$\begin{aligned}SS_{\text{Age at Intentional}} &= \sum_{j=1}^2 N_{j,\text{Intentional}} (\bar{X}_{j,\text{Intentional}} - \bar{X}_{\cdot,\text{Intentional}})^2 \\&= 10(19.3 - 15.65)^2 + 10(12.0 - 15.65)^2 = 266.45 \\df_{\text{Age at Intentional}} &= 1 \\MS_{\text{Age at Intentional}} &= 266.45 \\F_{\text{Age at Intentional}} &= 266.45/8.026 = 33.20\end{aligned}$$

In R (Nonstandard Error)

```
> simple.eysenck <- aov(recall~age,subset=(task=="Intentional"),data=data)
> summary(simple.eysenck)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
age	1	266.4	266.45	25.23	8.84e-05 ***
Residuals	18	190.1	10.56		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

A Function For You

```
simple.main.effects <- function(variable,factor1,factor2) {  
  # Extract MSE and error degrees of freedom from the full model  
  model.all <- aov(variable~factor1+factor2+factor1:factor2)  
  Error.df <- summary(model.all)[[1]][["Residuals","Df"]]  
  MSE <- summary(model.all)[[1]][["Residuals","Mean Sq"]]  
  
  # Extract the names of the factor levels  
  levels1 <- unique(factor1)  
  levels2 <- unique(factor2)  
  effects1 <- effects2 <- vector()  
  
  # Effects of factor2 when factor1 is fixed  
  for (i in 1:length(levels1)) {  
    model2 <- aov(variable~factor2,subset=(factor1==levels1[i]))  
    effects2[i] <- summary(model2)[[1]][["factor2","Mean Sq"]]  
    # Compute the F statistics and p-values for each level of factor1  
    F1 <- effects2/MSE  
    p1 <- 1 - pf(F1,length(levels1)-1,Error.df)  
  
    # Effects of factor1 when factor2 is fixed  
    for (i in 1:length(levels2)) {  
      model1 <- aov(variable~factor1,subset=(factor2==levels2[i]))  
      effects1[i] <- summary(model1)[[1]][["factor1","Mean Sq"]]  
    }  
    # Compute the F statistics and p-values for each level of factor2  
    F2 <- effects1/MSE  
    p2 <- 1 - pf(F2,length(levels2)-1,Error.df)  
  
    # Return the results. Don't forget to adjust your alpha level for  
    # multiple comparisons.  
    return(list("factor2"=cbind(levels1,F1,p1),"factor1"=cbind(levels2,F2,p2)))  
  }  
}
```

Function Output

```
> simple.main.effects(recall,task,age)
$factor2
      levels1      F1      p1
[1,] "Counting"  "0.155752457427662" "0.959934111598982"
[2,] "Rhyming"   "0.305274816558216" "0.873753620375231"
[3,] "Adjective" "8.99626194102174" "3.69923413590101e-06"
[4,] "Imagery"   "10.9898933960958" "2.6597628099978e-07"
[5,] "Intentional" "33.2001938252803" "0"

$factor1
      levels2      F2      p2
[1,] "Older"     "10.9500207669943" "0.00134718439777415"
[2,] "Younger"   "42.1690433337948" "4.48947046205461e-09"
```

Review

Remember:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$$

$$\eta^2 = \frac{SS_{\text{Between}}}{SS_{\text{Total}}}$$

Review

Remember:

$$SS_{\text{Total}} = SS_{\text{Cells}} + SS_{\text{Within}}$$

$$\eta_J^2 = \frac{SS_J}{SS_{\text{Total}}}$$

$$\eta_K^2 = \frac{SS_K}{SS_{\text{Total}}}$$

$$\eta_{J \times K}^2 = \frac{SS_{J \times K}}{SS_{\text{Total}}}$$

η^2

```
> library("effectsize")  
> eta_squared(eysenck, partial=FALSE)
```

```
# Effect Size for ANOVA (Type I)
```

Parameter	Eta2	95% CI
task	0.57	[0.45, 1.00]
age	0.09	[0.02, 1.00]
task:age	0.07	[0.00, 1.00]

Review

η^2 is biased: it overestimates the proportion of “explained” variance.

ω^2 is a correction to η^2 :

$$\omega^2 = \frac{SS_{\text{Between}} - (k - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}}$$

Review

η^2 is biased: it overestimates the proportion of “explained” variance.

ω^2 is a correction to η^2 :

$$\begin{aligned}\omega_{\text{Age}}^2 &= \frac{SS_{\text{Age}} - (2 - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}} \\ \omega_{\text{Task}}^2 &= \frac{SS_{\text{Task}} - (5 - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}} \\ \omega_{\text{Age} \times \text{Task}}^2 &= \frac{SS_{\text{Age} \times \text{Task}} - (5 - 1)(2 - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}}\end{aligned}$$

ω^2

```
> omega_squared(eysenck,partial=FALSE)
```

```
# Effect Size for ANOVA (Type I)
```

Parameter		Omega2		95% CI
task		0.55		[0.43, 1.00]
age		0.09		[0.02, 1.00]
task:age		0.06		[0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].

FAS in baby rats

Pregnant rats are given a diet that consists of 0% or 35% ethanol beginning at 5 or 15 days of gestation. After they are born, the young rats are trained to run a maze and the number of learning trials required is recorded.

Ethanol	Fetal Age	
	5 days	15 days
0%	5,4,3,4,2	6,7,5,8,4
35%	18,19,14,12,15	6,9,5,9,3

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Analyze the data using the following recommended steps:

1. Construct the cell means table and plot the means to determine the expected pattern of significant effects. Is there a main effect of Fetal Age? of Ethanol? Is there an interaction?
2. Compute the statistics necessary to compute the sums of squares. Use the computational forms.
3. Construct the ANOVA summary table and compute the F statistics. Find the critical F values for your α level in a table.
4. Compute η^2 and/or ω^2 for each main effect and interaction.